VF Asia Footwear Laboratory Testing Manual

Test Method for Lastometer	Test Method #: VF- B22 Page 1 of 3
Date Issued: May,2019 Revision #: 1 Reason for revision: Rectify the unit of measure from kg to kgf for strength of material	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To assess how much a material can be stretched simultaneously in two directions (distended) without being damaged. This method simulates conditions experienced during the lasting operation in footwear manufacture. This is mainly applicable to footwear upper materials but can be used to assess any sheet material such as leather, plastics, and textiles.

REFERENCES: SATRA TM 24, BS 3144: 1968

EQUIPMENTAND MATERIAL:

1. Lastometer meeting the requirement of SATRA TM 24 such as STD104 and STM463
2. Upper material such as Leather, synthetic , fabrics / textile , plastics

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.
2. Test specimens should be of sufficient size to allow them to be clamped firmly on the test machine so that they do not slip during the test.
3. For the SATRA lastometers the samples should be circles of 44.5mm diameter- for older models two semi-circular cut-outs 4 mm radius at opposite ends of a diameter are also necessary to hold the specimen on the two locating pins.
4. Samples must be taken from different areas of the test material, and at least 10cm from any edge. In the case of leather, follow sampling plan, VF - H02.

REPETITONS: Test three (3) samples per material.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Ensure the test machine is reset, with the central plunger retracted to zero or minimum distension. If it has a maximum force pointer turn it to zero.
2. Tightly clamp the specimen into the test machine, for thick specimens a very high clamping force will be necessary whereas with thin specimens care is required to avoid cutting into the specimen.
3. Start the test by forcing the plunger into the specimen at a rate of 0.20 ± 0.05 mm/s.

Test Method for Lastometer	Test Method #: VF- B22 Page 2 of 3
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For the SATRA Lastometer STD104, this involves turning the hand wheel clockwise at steady rate of about one revolution per second.

4. To record the point of first damage as the plunger stretches the specimen proceed based on the type of material such as leather, patent leather or synthetic material as indicated below:

i) Leather samples:

- When testing leathers the first damage usually occurs in the finish or the surface of the grain. Hence during the test continually observe the surface of the specimen, at the center where the maximum distension is taking place.
- At the first sign of surface cracking, record the force on the plunger and the distension of the specimen.
- However, if the grain is still intact, continue until grain crack occurs and record the distension and force at this point.

ii) Patent leather:

Patent leather is a special case because often the grain of the leather cracks before the patent surface film. Particular care is therefore required when observing patent leather test specimens.

- Two indications of the grain cracking are a small shallow depression in the surface of the film above the crack and a momentary drop in the force on the plunger.
- If the patent film cracks this should also be regarded as a “grain crack” whether or not a crack in the grain is visible.

iii) Synthetic materials

- When testing composite materials such as coated fabrics the first damage usually occurs within the material, with no visible damage at the surface. Hence during the test continually observe the force on the plunger as the specimen is stretched. When damage occurs to a layer of the specimen the force on the plunger either stops rising, or falls, however this may only be momentary. Record this value and the distension of the specimen at this point.
- If the bursting point is required then continue to observe the specimen and record the distension and force when the spherical ball appears through the sample.
- Retract the plunger and remove the test specimen. ***(See note below***)
- Inspect the marks on the top of the specimen left by the clamps. If there are signs of slippage occurring during the test, evident by a blurring of the inner clamping rings, or of tearing of the clamped edges; reject the results and repeat the procedure with a new specimen.

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***NOTE**: For removing the specimen after completion of the test - when using the SATRA lastometers, if removal of the specimen is difficult because the plunger has pierced it, the plunger can be lifted out of the machine together with the specimen. The plunger can then be manually pulled out of the specimen and replaced in the lastometer.*

CALCULATON:

Calculate the arithmetic mean of the results for the three specimens, quoting distension to the nearest 0.1mm, and force to the nearest 10 N (1 kgf).

REPORT:

1. Distension : mm
2. Force at first damage : kgf
3. For leather, grain crack if this does not occur at the first damage: kgf
4. Bursting point
5. Any deviations from this standard test method.